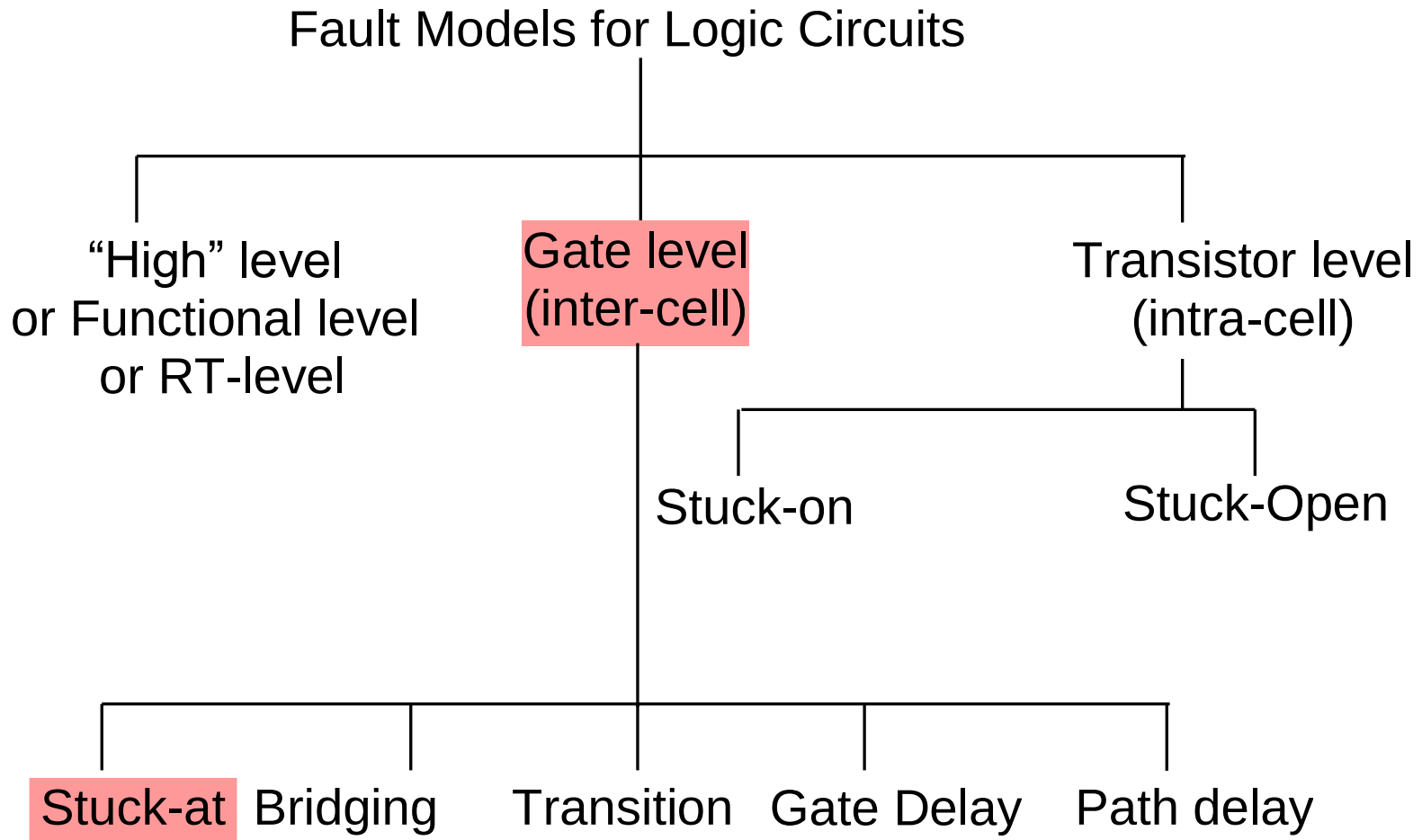


Fault Modeling

- Introduction
- Fault Models
 - ◆ Stuck-at fault (1961)
 - ◆ Bridging fault (1973)
 - ◆ Delay fault (1974)
 - ◆ Transistor level fault
- Fault Detection
- Fault Coverage
- Conclusion



Classification of Fault Models



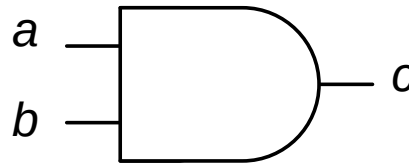
Single Stuck-at Fault Model

- **Single Stuck-at Fault (SSF) model [Galey 61]**
 - ♦ One signal line in Boolean network of logic gates is fixed to logic 0 or 1, independent of logic values on other signal lines
- Notations: **node x stuck-at fault**
 - * $x/0$, $x/1$,
 - * $x\ s@0$, $x\ s@1$
 - * $x\ SA0$, $x\ SA1$
- Number of faults is *linear* to circuit size
 - ♦ 2 faults (stuck-at one, stuck-at zero) per node
 - ♦ $2n$ SSF in a circuit of n nodes
- Most commonly studied fault model

SSF is Scalable for Large Circuits

Single Stuck-at Fault Example

- Example: SSF table of two input AND gate
 - ♦ Total **six** faults: **a/0, a/1, b/0, b/1, c/0, c/1**
- Test set $ab = \{01, 11, 10\}$
 - ♦ **100%** SSF fault coverage, detects all SSF
- Minimum test length = **3** for 100% SSF fault coverage



Input <i>a b</i>	Fault-free Output	Faulty Output Value with SSF					
		<i>a/0</i>	<i>a/1</i>	<i>b/0</i>	<i>b/1</i>	<i>c/0</i>	<i>c/1</i>
0 0	0	0	0	0	0	0	<u>1</u>
0 1	0	0	<u>1</u>	0	0	0	<u>1</u>
1 1	1	<u>0</u>	1	<u>0</u>	1	<u>0</u>	1
1 0	0	0	0	0	<u>1</u>	0	<u>1</u>

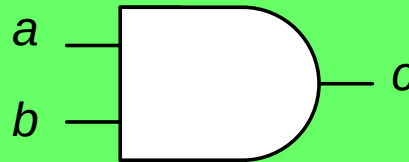
*erroneous output values highlighted

Quiz (Revisited)

- (Cont'd from P.3) Manger asked you to pick 2 patterns...

**Q: Based on SSF model, which patterns do you pick ?
What is maximum fault coverage?**

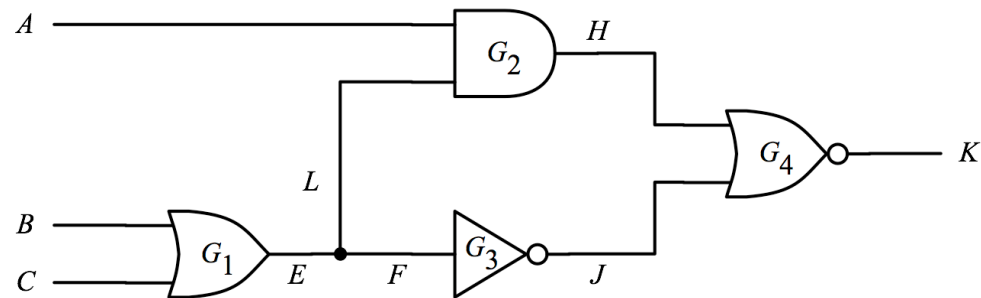
A:



Input <i>a b</i>	Fault-free Output	Faulty Output Value with SSF					
		<i>a/0</i>	<i>a/1</i>	<i>b/0</i>	<i>b/1</i>	<i>c/0</i>	<i>c/1</i>
0 0	0	0	0	0	0	0	<u>1</u>
0 1	0	0	<u>1</u>	0	0	0	<u>1</u>
1 1	1	<u>0</u>	1	<u>0</u>	1	<u>0</u>	1
1 0	0	0	0	0	<u>1</u>	0	<u>1</u>

Fanout Stems and Branches

- SSF on fanout stem *not equivalent to* SSF on fanout branches
 - ♦ Faults on stems and faults on branches are **counted separately**
- Example: *E* is fanout stem; *L, F* are fanout branches
 - ♦ 6 faults are not equivalent



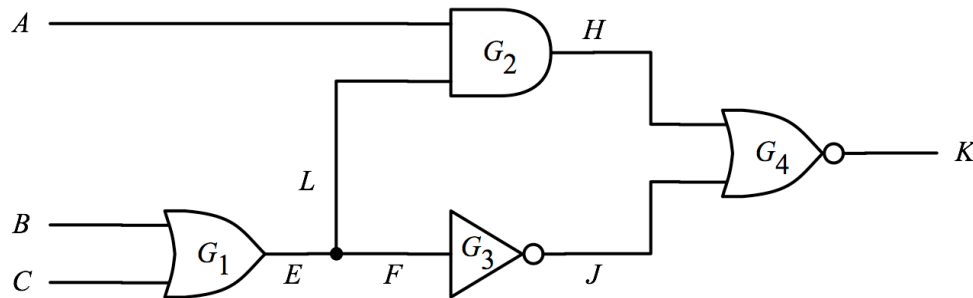
Input			Output						
A	B	C	good	E/0	F/0	L/0	E/1	F/1	L/1
0	0	0	0	0	0	0	1	1	0
0	0	1	1	0	0	1	1	1	1
0	1	0	1	0	0	1	1	1	1
0	1	1	1	0	0	1	1	1	1
1	0	0	0	0	0	0	0	1	0
1	0	1	0	0	0	1	0	0	0
1	1	0	0	0	0	1	0	0	0
1	1	1	0	0	0	1	0	0	0

**Stem Faults $\neq$
Branch Faults**

Quiz

Q: How many single stuck-at faults in this circuit?

- A. 8 fault**
- B. 14 faults**
- C. 18 faults**



Multiple Stuck-at Faults

- **Multiple stuck-at fault (MSF) model**

- ♦ More than one stuck fault in a circuit

- Quiz

Q: how many multiple stuck-at faults in a circuit of n nodes?
(do not include single stuck-at fault)

A:

- Too many MSF, not scalable for large circuits

- ♦ Fortunately, SSF test set can detect **many** MSF

MSF Not Considered in Practice

Simulation Results [Hughes 86]

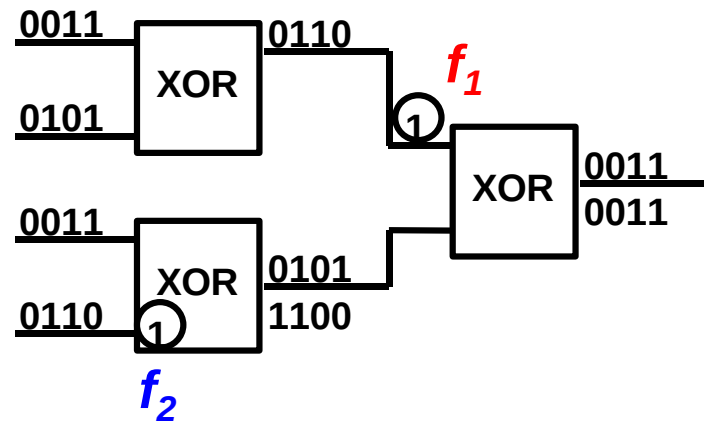
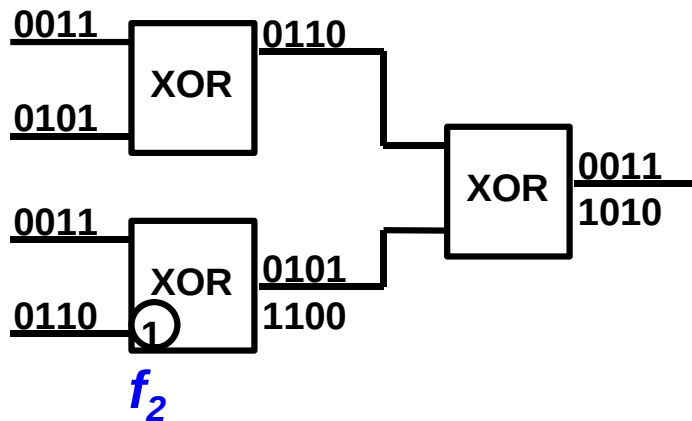
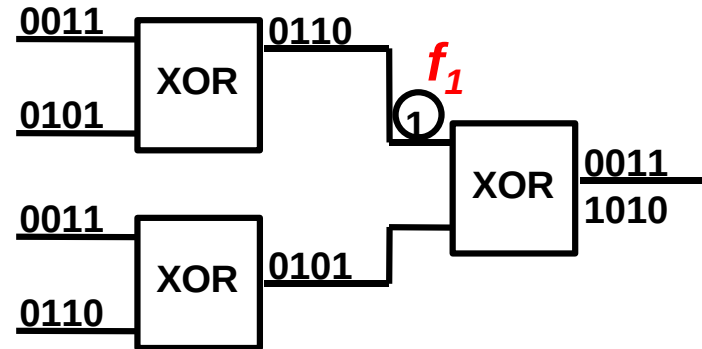
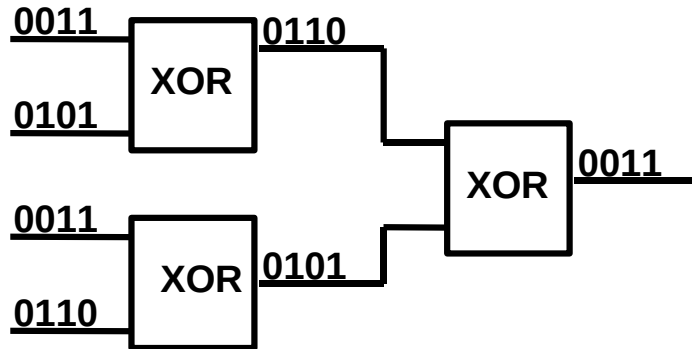
- How effective is SSF test sets for multiple stuck-at faults?
 - ♦ 14-input ALU 74LS181 circuit
 - ♦ 400 single stuck-at faults
 - ♦ 79,600 double stuck-at faults
 - ♦ 16 different test sets

Test Set	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Length	12	12	12	12	12	12	12	14	14	14	14	17	35	124	135	352
Undetected Double-stuck faults	9	8	1	9	28	13	19	4	14	11	3	30	0	0	0	0

- Observation #1: Some shorter tests are better than longer tests
 - ♦ Smart test generation/selection is important
- Observation #2: Most double SA faults are detected by SSF test sets
 - ♦ But some exceptions (see next slide)

Why DSF NOT Detected?

- Example: double stuck-at fault not detected by 100% SSF test set



No fault detected

Fault Masking

- Fault f_2 **masks** fault f_1 if
 - ♦ A test for f_1 fails to detect f_1 in the presence of f_2
- When a test fails to detect a fault it is supposed to detect
 - ♦ We call it **test invalidation**
- Fault masking **rarely** happens in practice
 - ♦ Trouble more for **diagnosis** than for testing

Fault Masking Invalidates Tests
Fault Masking Makes Diagnosis Difficult

Fault Modeling

- Fault Models

- ◆ Stuck-at fault

- * Number of single stuck-at faults is linear to circuit size : $2n$
- * Multiple stuck-at faults are exponential in number
- * Fault masking means two faults cancel each other



FFT

- Fault masking degrades test effectiveness
 - ♦ does fault masking degrade SSF coverage?

